📚 STUDY GUIDE – SCIENCE

Year 2 · Unit 3 – Position and Motion · Test (AF): November 4th

📍 PART 1 – POSITION: WHERE IS IT?

1. Prepositions of place

Prepositions of place tell us **where** something is.

Word	What it means	Example
in / inside 📦	in the middle of something	The teddy bear is **in** the box. The fish is **inside** the bowl.
on 📕	on top, **touching** it	The apple is **on** the book.
under ⚽	lower, right below it	The ball is **under** the table.
below	lower than something	The cat is **below** the shelf.
above / over 💡	higher, **not touching**	The lamp is **above** the table.
behind 🌳	at the back of it	The boy is **behind** the tree.
in front of 🐶	at the front of it	The dog is **in front of** the chair.
between	in the middle of **two** things	The teddy bear is **between** the two plants.
next to ⏰	side by side	The clock is **next to** the lamp.
near 🏠	close, just a little distance	The tree is **near** the house.
far from ↔️	a long distance away	The tree is **far from** the house.

⚠️ **on** × **above**: **on** is touching. **above** is higher, but **not touching**.

⚠️ **near** × **far from**: near = close 👣. far from = a long way 🚗.

⚠️ **between** always needs **two** things: one on each side.

⭐ **To remember:** Ask "**Where is it?**" The answer uses a preposition: "It is **under** the table."

2. Directions: left and right

Directions help us say **where** to go or **where** something is.

- ⬅️ **left** – Hold up your hands. Your **left** hand makes the letter **L** with the thumb and finger.
- ➡️ **right** – the other side.
- We also use **between**, **under**, **next to**... to give directions.

3. Changing perspectives 👀

We can see things from **different positions**. What we see changes depending on **where** we are looking from.

Eiffel Tower seen from the...	Where am I?
front	standing on the grass, looking straight at it
side	looking at it from one side (by the river)
top ⬇️	high up, looking **down**
bottom ⬆️	right under it, looking **up**

- One wooden shape can make a **triangle**, a **square** and a **circle** shadow. Same object, different views!
- The giraffe and the woman paint the same thing, but each one sees it in a different way. Nobody is wrong: they just have a **different view**.

⭐ **To remember:** Same thing + different position = different view.

4. Reading maps 🗺️📍

- A **map** shows places and the **path (way)** to get there.
- There are maps of cities, of the **school**, and maps on the phone (GPS).
- Pay **a lot of attention** to the **details** and to the **path**.
- If you **run** or **don't** pay attention, you might get **lost**!

🏃 PART 2 – MOVEMENT

5. Movements of objects

Movement	What it means	Examples
bounce 🏀	goes **up and down** after hitting a surface	soccer ball, tennis ball, basketball
fly ✈️	moves **through the air**	airplane, hot air balloon, kite
roll 🛞	moves by **turning over and over**	wheel, can, beach ball
slide 🛝	moves **smoothly along** a surface	slide, ice cube, book on a table
spin 🌀	turns **around and around**	pinwheel, fan, top
swing	moves **back and forth**, like a pendulum	swing, hanging plant, door

⚠️ **roll** × **spin**: **roll** = turns and **goes** to another place. **spin** = turns but **stays** in the same place.

⚠️ Some objects do more than one movement: a ball can **roll** and **bounce**.

Things also move in different **directions** and at different **speeds**:

Direction ↔️ ↕️ 🔄	Speed
back and forth (a swing) **up and down** (a bouncing ball) **round and round** (a fan)	**slow** 🐢 **fast** 🏎️

⭐ **To remember:** "**How does it move?**" Say the **way** (roll, slide...), the **direction** (up and down...) and the **speed** (slow or fast).

6. Movements of my body ✋

With my body	With my hand
swim, climb, play, pick, walk, sit	touch, lift, point, write, grab, push

7. Movement of animals 🐾

Animals move in different ways:

Animal	How it moves
fish 🐟	**swims**
bird 🐦	**flies**
kangaroo 🦘 / frog 🐸	**hops / jumps**
snake 🐍	**slithers** (moves on its belly)
horse 🐴 / dog 🐕	**walks / runs**

💪 PART 3 – PUSH AND PULL

8. Push and pull

A **force** is a **push** or a **pull**. It makes things move from one place to another. Push and pull are **opposite** forces.

PUSH ➡️ away from me	PULL ⬅️ closer to me
riding a bike (pushing the pedals) 🚲 pushing a box 📦 pushing a door 🚪 kicking a ball ⚽ moving a shopping cart 🛒 hitting a volleyball 🏐 pressing the piano keys 🎹	tug of war 🪢 pulling a wagon opening a drawer catching a fish 🎣 pulling the kite string 🪁 picking fruit from the plant 🍓

⭐ **To remember:** Ask "**Is it going away from me or coming to me?**"
Away from me = **PUSH**. Closer to me = **PULL**.

🌬️💧 PART 4 – WIND AND WATER

9. Natural forces: helpful and harmful

Wind is **moving air**: a gentle breeze or a strong gust. **Wind** and **water** are **natural forces**. They can be **helpful** 🙂 or **harmful** 😕 to us.

	Helpful 🙂	**Harmful 😕**
Water 💧	waters the plants and the farms 🌱 makes energy (dam) ⚡ carries boats and ships 🚢 we drink, cook and wash 🚿 rivers are home to fish, frogs and beavers	tsunami 🌊 heavy rain and floods
Wind 🌬️	makes energy (wind turbines) spreads the seeds (dandelion) pushes sailboats ⛵ flies kites 🪁	tornado 🌪️ strong winds

⭐ **To remember:** The **same** force can help or hurt. A little wind spreads seeds; very strong wind can break trees and houses.

10. Wind and water change the land 🏜️🏞️

Wind and water change the land **slowly**, over **many, many years**.

Wind 🌬️	**Water 💧**
moves sand in the desert → **sand dunes** blows sand on rocks (like sandpaper) → **rock arches**	fast **rivers** wear the rock → **valleys** and **canyons** **rain** carries soil → changes the hills

⭐ **Fun facts:** One wind turbine can make energy for about **1,000 houses**! The wind in a **tornado** is faster than a race car.

📝 PRACTICE TEST – SCIENCE

Year 2 · Unit 3 – Position and Motion

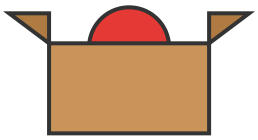
Name: _____

Date: ___ / ___ / _____

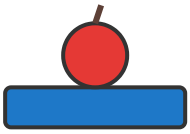
PART 1 – WHERE IS IT?

1. Look at the pictures. Write the right word.

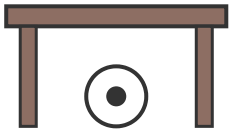
in	on	under	above



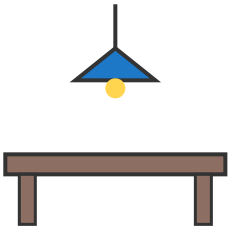
a) _____



b) _____



c) _____



d) _____

2. Complete with the words in the box. Use each word once.

behind	between	in front of	next to	near	far from

a) The boy is hiding _____ the tree. We can't see him!

b) The teddy bear is _____ the two plants.

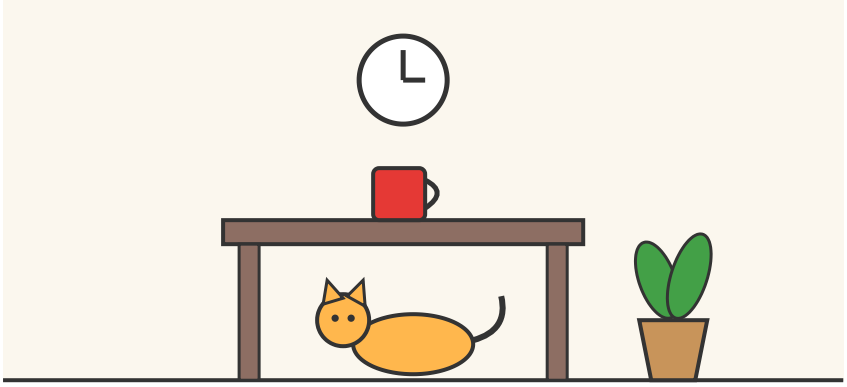
c) The dog is _____ the chair. I see the dog first.

d) The clock is _____ the lamp, side by side.

e) Grandma's house is _____ my house. We need a car to go there.

f) The park is _____ my house. I can walk there in two minutes.

3. Look at the picture. Write **T** (true) or **F** (false).



a) The cup is **on** the table.	
b) The cat is **on** the table.	
c) The plant is **next to** the table.	
d) The clock is **under** the table.	

4. The lamp is high over the table. It is **not touching** the table. The lamp is _____ the table.

a) on

b) above

c) in

5. Look at the pictures from left to right.



a) What is to the **left** of the star? _____

b) What is to the **right** of the star? _____

PART 2 – CHANGING PERSPECTIVES

6. What we see changes depending on...

a) the weather.

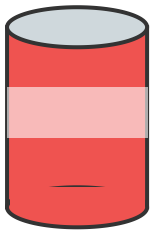
b) where we are looking from.

c) the color of the object.

7. Match. Where am I looking at the Eiffel Tower from?

I stand on the grass and look straight at it. •	• bottom
I fly in a helicopter and look down. •	• front
I lie under the tower and look up. •	• side
I am on a boat on the river, beside it. •	• top

Look at this can.



8. If you look at the can from the **top**, what shape do you see?

a) a circle ⚬

b) a triangle 🔺

c) a star ⭐

9. If you look at the can from the **side**, what shape do you see?

a) a circle

b) a rectangle

c) a triangle

10. Ana looks at a shape from the **front**. Leo looks at the same shape from the **top**. They see different shapes. Who is right?

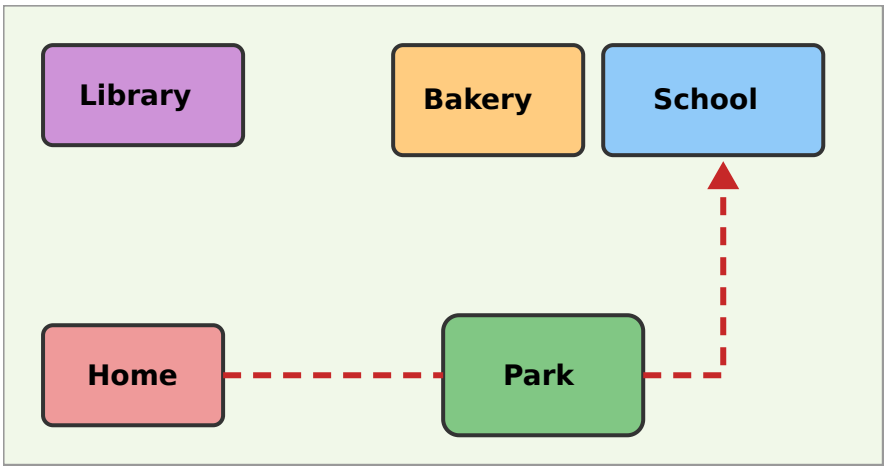
a) Only Ana.

b) Only Leo.

c) Both! They are looking from different positions.

PART 3 – READING MAPS

Look at the map. Follow the dashed path from Home.



11. Which place do you pass **before** you get to the School?

a) the Library

b) the Park

c) the Bakery

12. What is **next to** the School? _____

13. Mark (✓) the good tips for reading a map.

☐ Pay attention to the details.

☐ Run very fast.

☐ Follow the path (way).

☐ Don't look at the map.

PART 4 – MOVEMENT OF OBJECTS

14. Match the movement to what it means.

bounce •	• moves through the air
fly •	• turns around and around
roll •	• goes up and down after hitting a surface
slide •	• moves back and forth, like a pendulum
spin •	• moves by turning over and over
swing •	• moves smoothly along a surface

15. Complete with the words in the box.

bounce	fly	roll	slide	spin	swing

a) A kite can _____ in the sky.

b) A car wheel can _____ on the road.

c) A top can _____ round and round.

d) A tennis ball can _____ on the floor.

e) A door can _____ open and closed.

f) An ice cube can _____ on the table.

16. A ball can do more than one movement! Write **two** movements a ball can do.

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17. Direction and speed. Complete with the words in the box.

back and forth	up and down	round and round

a) A swing moves _____ .

b) A bouncing ball goes _____ .

c) A fan spins _____ .

d) A snail 🐌 moves **slow** / **fast**. (circle)

e) A race car 🏎️ moves **slow** / **fast**. (circle)

PART 5 – MY BODY AND THE ANIMALS

18. Mark (✓) the movements you can do with your **hand**.

☐ touch
☐ swim
☐ grab
☐ walk
☐ write
☐ point

19. Match the animal to how it moves.

fish 🐟 •	• slithers
bird 🐦 •	• flies
kangaroo 🦘 •	• swims
snake 🐍 •	• hops

PART 6 – PUSH AND PULL

20. Write **PUSH** or **PULL**.

a) kick a ball ⚽ _____

b) open a drawer _____

c) move a shopping cart 🛒 _____

d) catch a fish 🎣 _____

e) press the piano keys 🎹 _____

f) close a door 🚪 _____

g) play tug of war 🪢 _____

h) pick an apple from the tree 🍎 _____

i) ride a bike 🚲 _____

j) hit a volleyball 🏐 _____

21. Complete.

A **push** moves something _____ from me.

A **pull** moves something _____ to me.

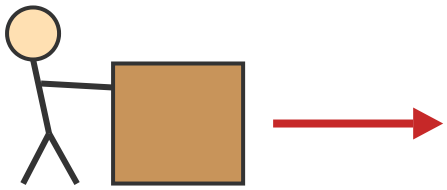
22. Push and pull are two kinds of...

a) colors.

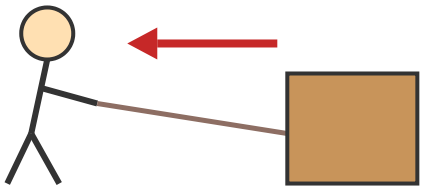
b) forces.

c) animals.

23. The arrow shows where the box moves. Write **PUSH** or **PULL**.



a) _____



b) _____

PART 7 – WIND AND WATER

24. Write 🙂 for **helpful** or 🙁 for **harmful**.

( ) water the plants

( ) a tsunami

( ) wind turbines make energy

( ) a tornado

( ) wind spreads the seeds

( ) a big flood

( ) a dam makes energy with water

( ) strong winds break the trees

25. Wind and water are _____ forces.

a) natural

b) pretend

c) made in a factory

26. How can **wind** help us?

a) It makes floods.

b) It makes energy with wind turbines.

c) It breaks houses.

27. Write **one** way **water** can be harmful.

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28. Complete with the words in the box.

moving air	**sand dunes**	**canyon**	**sailboat**

a) Wind is _____ .

b) The wind moves the sand in the desert and makes _____ .

c) A fast river wears the rock and makes a deep _____ .

d) The wind fills the sails and pushes a _____ .

🌟 FINAL CHALLENGE

A Windy Day in the Park

It is a windy day. Mia and her brother Tom are in the park. Mia's kite flies high above the trees. Mia pulls the string to bring the kite closer.

Tom kicks his ball. The ball rolls under a bench! Then Tom sits on the swing, and Mia pushes him. The swing goes back and forth.

29. Answer about the story.

a) Where is the kite? It is _____ the trees.

b) Mia **pushes** / **pulls** the string. (circle)

c) Tom kicks the ball. Is it a push or a pull? _____

d) How does the ball move? It _____ . Where is it now? _____

e) Mia looks **up** at the kite. She sees it from the **top** / **bottom**. (circle)

f) Is the wind helpful or harmful today? Why?

✅ ANSWER KEY (GABARITO)

1. a) in · b) on · c) under · d) above
2. a) behind · b) between · c) in front of · d) next to · e) far from · f) near
3. a) T · b) F (it is under) · c) T · d) F (it is above)
4. b) above
5. a) the apple · b) the heart
6. b) where we are looking from
7. grass/straight → front · helicopter/down → top · under/up → bottom · boat/beside → side
8. a) a circle
9. b) a rectangle
10. c) Both
11. b) the Park
12. the Bakery
13. ✓ Pay attention to the details · ✓ Follow the path
14. bounce → up and down after hitting a surface · fly → through the air · roll → turning over and over · slide → smoothly along a surface · spin → around and around · swing → back and forth
15. a) fly · b) roll · c) spin · d) bounce · e) swing · f) slide
16. roll and bounce (fly also accepted)
17. a) back and forth · b) up and down · c) round and round · d) slow · e) fast
18. ✓ touch · ✓ grab · ✓ write · ✓ point
19. fish → swims · bird → flies · kangaroo → hops · snake → slithers
20. a) PUSH · b) PULL · c) PUSH · d) PULL · e) PUSH · f) PUSH · g) PULL · h) PULL · i) PUSH · j) PUSH
21. away · closer
22. b) forces
23. a) PUSH · b) PULL
24. 🙂 plants · 😕 tsunami · 🙂 turbines · 😕 tornado · 🙂 seeds · 😕 flood · 🙂 dam · 😕 strong winds
25. a) natural
26. b) energy with wind turbines
27. tsunami, heavy rain or floods
28. a) moving air · b) sand dunes · c) canyon · d) sailboat
29. a) above · b) pulls · c) push · d) rolls; under the bench · e) bottom · f) helpful – it makes the kite fly

💡 Para estudar com ela (prova na quarta, 04/11)

Qui 29/10 – Preposições (itens 1–2). Brincadeira "Onde está?": esconda um bichinho de pelúcia pela casa; ela acha e responde em inglês com a preposição (*It is under the bed*). Inclua *left/right*.

Sex 30/10 – Perspectivas e mapas (itens 3–4). Mostre uma caneca de cima, de lado e de baixo e peça que ela desenhe o que vê. Depois, desenhem juntos um mapa simples da casa com um caminho tracejado.

Sáb 31/10 – Movimentos (itens 5–7). Caça aos objetos: achar em casa algo que *rolls, bounces, spins, slides, swings*. Mímica de animais (*swim, fly, hop, slither*). Para cada objeto, ela diz também a direção (*back and forth, up and down, round and round*) e se é *slow* ou *fast*.

Dom 01/11 – Push e pull (item 8). Ela faz ações (abrir gaveta, fechar porta, chutar bola) e diz "push" ou "pull", usando a pergunta: "vai para longe de mim ou vem para perto?". Inclua os exemplos dos vídeos: tocar piano e carrinho de supermercado (*push*); puxar a linha da pipa e colher uma fruta (*pull*).

Seg 02/11 – Vento e água (itens 9–10). Separar as imagens do slide em 🙂 e 😕. Soprar bolhas de sabão para mostrar que vento é ar em movimento. Em uma bandeja com areia, soprar com um canudo para formar "dunas" e despejar um fio de água para cavar um "cânion".

Ter 03/11 – Simulado sem consultar o resumo; corrigir juntos e refazer só o que errou.

Pontos que mais confundem: *on* × *above* (encosta ou não); *near* × *next to*; *roll* × *spin*; fechar porta e andar de bicicleta = *push*; o mesmo vento ou a mesma água pode ser *helpful* ou *harmful*; dunas = vento, cânion e vale = rio.